

Hate Speech Early-Warning & Legal Triage System

A decision-support system for Gilgit-Baltistan that identifies potentially unlawful online content, maps it to the applicable law, and places every case before a human officer.

SUBMITTED TO

The Home Secretary
Home & Prison Department
Government of Gilgit-
Baltistan

SUBMITTED BY

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DATE

16 Sep 2026

REFERENCE

NA

YEAR 1 PROGRAM

PKR 29,894,666

Three gaps in current capability

Latency

Content that incites sectarian hostility circulates widely before it is noticed. By the time a complaint arrives, the administration's options have narrowed.

Language

Much local discourse is Roman Urdu or regional — Shina, Balti, Burushaski. General-purpose commercial moderation tools handle these poorly or not at all.

Evidence

When a case is escalated, its legal basis is assembled manually and inconsistently. Officers need the applicable provision identified alongside the content.

A fourth consideration governs the design. In a region with real sectarian sensitivity, **a system that produces false accusations is worse than no system at all.**

Six agents. One terminus.

Each component performs one task and passes an enriched case record onward.
Processing ends, in every case, at human review.



Dashed nodes fire conditionally on confidence and risk tier. The terminus does not.

The system cannot act on its own

This is not a policy commitment that could be relaxed later. It is enforced in the data layer.

- **Enforced at the database**

A constraint rejects any case record that attempts to bypass human review. There is no code path and no configuration setting that can disable it.

- **Advisory legal output**

Statutory mapping is a drafting aid for a legally trained reviewer to confirm — never a determination. The interface states this on every case.

- **No fabricated citations**

Where a provision cannot be verified, the system records "not yet mapped" rather than proposing one.

- **Full audit trail**

Every decision records the officer, timestamp, decision and written reasoning — available for departmental or judicial scrutiny.

Accountability boundary

```
requires_human_review
```

```
INTEGER NOT NULL
```

```
DEFAULT 1
```

```
CHECK (requires_human_review = 1)
```

Every case file carries this constraint. A record that tried to skip review would be rejected by the database itself.

Flagged content, matched to statute

INSTRUMENT	PROVISION	SUBJECT
Pakistan Penal Code	153-A	Promoting enmity between groups
Pakistan Penal Code	295-A	Deliberate acts to outrage religious feelings
PECA 2016	Cyber-offence sections	Offences via electronic communication
Constitution of Pakistan	Article 20	Freedom of religion — recorded as the competing right to be weighed
Gilgit-Baltistan local law	Not yet mapped	Stated explicitly rather than guessed

Article 20 is carried deliberately. A system that only ever cites the offence, and never the right being weighed against it, is an enforcement tool rather than a legal one.

What exists today, stated plainly

A working prototype is built and running. Presenting it as a finished system would be a disservice — and in a security context, a dangerous one.

COMPONENT	STATUS	NOTE
Classification model	OPERATIONAL	Not yet validated on GB content
Agent orchestration	OPERATIONAL	Tested end-to-end
Human review queue	OPERATIONAL	Constraint enforced at database level
Legal mapping	NEEDS VALIDATION	Requires advocate sign-off
Campaign clustering	SIMPLIFIED	Embeddings + cosine, not full ULTRA
Sarcasm detection	PLACEHOLDER	Rule-based; not a trained model
District geo-tagging	UNAVAILABLE	No reliable location source
Authentication & RBAC	NOT BUILT	Required before production

The published **95.7%** accuracy figure was obtained on the model's original academic test set. It has **not** been validated on Gilgit-Baltistan content, and real-world accuracy on local dialects should be expected to be lower until retraining is complete. No accuracy guarantee is offered on that basis.

One year. One approved figure.

A single twelve-month engagement covering all development, hardware and twelve months of subscriptions — with an early review point at Month 4, before the greater part of the money is committed.

Stage 1 · Months 1-3

Validation

Gilgit & Skardu. Delivers a measured accuracy report on the Department's own content. PKR 6.93M.

Stage 2 · Months 4-12

Build & rollout

Model retraining, security hardening, authentication, remaining districts, officer training. PKR 22.96M.

Year 2 onward

Not sought here

Recurring charges shown for transparency only, to be submitted separately when lower-cost options may exist.

The two stages are **one approved figure of PKR 29,894,666**, not two purchases. Stage 1 is the first three months of it.

Stage 1 — Validation Pilot

COST HEAD	PKR
Human resources — 7 roles, part-time over 3 months	3,497,500
Hardware — GPU workstation, UPS, 2 review stations	1,580,000
Software & subscriptions (3 months)	227,160
Data & annotation — initial GB sample set	350,000
Training — single session, 2 districts	150,000
Travel & field	250,000
Administrative overhead (6%)	363,280
Contingency (8%)	513,435
Total — Stage 1	6,931,375

6.93M

PKR — within the Year 1 total
≈ USD 24,755

3

Months to a measured result
Gilgit & Skardu

23%

Of the Year 1 program
Review point at Month 4

Year 1 Complete Program

COST HEAD	PKR
Human resources — 11 roles, part-time over 12 months	18,290,000
Hardware (CapEx) — single server, UPS, 4 workstations	3,210,000
Software & subscriptions — 12 months included	1,313,440
Data & annotation — ~25,000 GB samples	1,200,000
Training & capacity building	800,000
Legal & governance	600,000
Travel & field	700,000
Administrative overhead (6%)	1,566,806
Contingency & risk reserve (8%)	2,214,420
Total — Year 1 (all-inclusive)	29,894,666

29.89M

PKR — under the approved ceiling
≈ USD 106,767

12

Months of subscriptions included
No further Year 1 cost

~100K

Posts / month capacity
Scale tier, single server

Recurring charges — not sought here

Shown for planning transparency only. Per the Department's direction these form a separate submission nearer the time, when lower-cost technical options may be available.

ITEM	PKR / YEAR
Skeleton operations team	6,330,000
Apify subscription & overage	1,004,640
Monitoring & off-site backup	268,800
Hardware maintenance (8%)	256,800
Power & connectivity	600,000
Annual security review	400,000
Model retraining, twice yearly	800,000
Contingency (10%)	966,024
Indicative annual total	10,626,264

72.40M

PKR — indicative five-year total

≈ USD 258,570 · if later years approved

Retraining is not discretionary maintenance. Without it the model grows progressively less reliable while continuing to *appear* authoritative — the most dangerous failure mode for a system of this kind.

What the envelope buys — and what it does not

Delivering within the approved ceiling required real reductions in scope, not a discount on the same scope. Stating them plainly matters: an approval granted on an unstated assumption of wider scope fails during delivery.

CAPABILITY	YEAR 1	DEFERRED
Core pipeline & officer dashboard	FULL	—
Human review, audit trail, accountability constraint	FULL	—
Legal mapping with advocate validation	FULL	—
Server infrastructure	SINGLE SERVER	High-availability pair
Disaster recovery site	NOT INCLUDED	Off-site backup only
Ingestion capacity	~100K / MONTH	300–500K (Business tier)
GB annotation corpus	~25,000 SAMPLES	100,000 samples
Security assurance	ONE PEN TEST	ISO 27001, bias audit
Officer workstations	4	15

None of these reductions touch the safeguards. Human review, the audit trail, the accountability constraint and legal validation are funded in full — they are what make the system defensible if a decision taken with its assistance is challenged.

Principal risks & mitigation

RISK	SEVERITY	MITIGATION
False positive leads to wrongful action against a citizen	HIGH	Mandatory human review; no autonomous path exists; legal officer sign-off
Under-performance on Shina / Balti / Burushaski	HIGH	GB corpus collection and retraining funded; per-language accuracy reported
Perceived as mass surveillance	HIGH	Published transparency policy, grievance mechanism, independent oversight
Single-server failure causes outage	MED-HIGH	Accepted consequence of the budget envelope — mitigated by spares, nightly off-site backup and a documented rebuild procedure, not automatic failover
Volume exceeds Scale-tier capacity	MEDIUM	Volume monitoring with alerting; tier upgrade costed and presented before any breach
Platform or provider terms change	MEDIUM	Pluggable ingestion; licensed provider only; no direct scraping
Foreign-exchange exposure	MEDIUM	FX contingency in reserve; annual re-basing of USD items
Key-person dependency	MEDIUM	Documentation, train-the-trainer, departmental capability transfer

Approve PKR 29,894,666

The Year 1 Complete Program — all development, hardware and twelve months of subscription charges, within the ceiling already approved by the Home Secretary.

Costed to the envelope, honestly

Scope was reduced in infrastructure resilience, assurance depth and capacity — and stated plainly, so nothing fails by surprise during delivery.

Safeguards funded in full

Human review, audit trail, the accountability constraint and legal validation are untouched. They are what make the system defensible.

A review point at Month 4

Stage 1 delivers measured accuracy on GB content before the greater part of the expenditure is committed — with a documented basis to continue, adjust or halt.

SIGNATURE

Zain Abbas

TECHNICAL LEAD, ZIRCON LOGICS

DATE